

Formula:

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

site (x)	price (y)	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$
8	10	-2	-3	4
10	13	0	0	0
12	16	2	3	4

$$\bar{x} = \frac{8+10+12}{3} = 10$$

$$\bar{y} = \frac{10+13+16}{3} = 13$$

$$\therefore m = \frac{6 + 0 + 6}{8} = 1.5$$

$$c = \bar{y} - m\bar{x}$$

$$= 13 - 1.5 \times 10$$

$$= 13 - 15$$

$$= -2$$

$$\therefore y = 1.5x - 2 \quad \checkmark$$

y = Real

$\hat{y}$ = Predicted

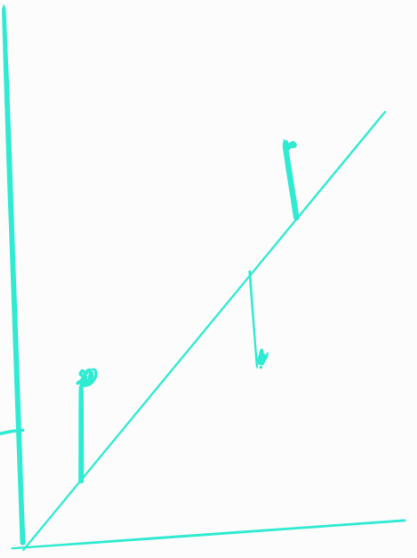
$$\text{Error} = \sum (y_i - \hat{y})$$

Derivation:

$$\text{Error} = \sum (y_i - \hat{y})$$

Square error:

$$\sum (y_i - (mx_i + c))^2$$



$$\therefore f(m, c) = \sum (y_i - (mx_i + c))^2$$

$$\frac{\partial}{\partial c} f(m, c) = \frac{\partial}{\partial c} \sum (y_i - (mx_i + c))^2$$

$$\Rightarrow \frac{\partial}{\partial c} \sum (y_i - mx_i - c)^2$$

$$\Rightarrow -2 \sum (y_i - mx_i - c)$$

Setting the derivative equal to zero:

$$-2 \sum (y_i - mx_i - c) = 0$$

$$\sum (y_i - mx_i - c) = 0$$

$$\sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - nc = 0$$

$$\Rightarrow nc = \sum y_i - m \sum x_i$$

$$\Rightarrow c = \frac{\sum y_i - m \sum x_i}{n}$$

$$\Rightarrow c = \frac{\sum y_i}{n} - \frac{m \sum x_i}{n}$$

$$c = \bar{y} - m \bar{x}$$

Partial derivative respect to m

$$\frac{\partial}{\partial m} f(m, c) = \frac{\partial}{\partial m} \sum (y_i - (mx_i + c))^2$$

$$\Rightarrow \frac{\partial}{\partial m} \sum (y_i - mx_i - (\bar{y} - m\bar{x}))^2$$

$$\Rightarrow \frac{\partial}{\partial m} \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\Rightarrow 2 \sum (y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\Rightarrow \sum ((y_i - \bar{y}) - m(x_i - \bar{x}))(\bar{x} - x_i) = 0$$

$$\Rightarrow \sum (y_i - \bar{y})(\bar{x} - x_i) - m \sum (x_i - \bar{x})^2 = 0$$

$$\Rightarrow \sum (y_i - \bar{y})(\bar{x} - x_i) - m \sum (x_i - \bar{x})^2 = 0$$

$$\Rightarrow m \sum (x_i - \bar{x})^2 = \sum (y_i - \bar{y})(x_i - \bar{x})$$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

$$\text{Total Error} = \sum (y_i - \hat{y}_i)$$

$$\text{Loss function} = \frac{1}{2n} \sum_{i=1}^n (y_i - (mx_i + c))^2$$

$$m := m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$c := c - \alpha \frac{\partial}{\partial c} J(m, c)$$

$$\frac{\partial}{\partial m} J(m, c) = \frac{\partial}{\partial m} \frac{1}{2n} \sum (y_i - (mx_i + c))^2$$

$$\Rightarrow -\frac{1}{n} \sum (y_i - (mx_i + c))^2 x_i \quad \checkmark$$

$$\frac{\partial}{\partial c} J(m, c) = \frac{\partial}{\partial c} \frac{1}{2n} \sum (y_i - (mn_i + c))^2$$

$$\Rightarrow -\frac{1}{n} \sum (y_i - (mn_i + c)) \quad \checkmark$$

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}) n_i$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y})$$

x	y	$\hat{y}_1$	$\hat{y}_2$	$\hat{y}_3$	
1	1	0	0.1		
2	2	0	0.175		
3	3	0	0.25		
4	0	0	0.325		

Initially,

$$m = 0$$

$$c = 0$$

$$\alpha = 0.01$$

now,

$$m := m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$c := c - \alpha \frac{\partial}{\partial c} J(m, c)$$

Here,

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}_i) x_i$$

$$\Rightarrow -\frac{1}{4} \left\{ (1-0) \times 1 + (2-0) \times 2 + (3-0) \times 3 + (4-0) \times 4 \right\}$$

$$\Rightarrow -\frac{1}{4} \times (1 + 4 + 9 + 16)$$

$$\Rightarrow -7.5$$

$$\therefore m := 0 - 0.01 \times (-7.5) = 0.075 \quad \checkmark$$

again:

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y})$$

$$\Rightarrow -\frac{1}{4} \sum (1-0) + (2-0) + (3-0) + (4-0) \}$$

$$= -\frac{1}{4} (1+2+3+4)$$

$$= -2.5$$

$$\text{So, } c_1 = c - \alpha \frac{\partial}{\partial c} J(m, c)$$

$$= 0 - 0.01 (-2.5)$$

$$= 0.025 \quad \checkmark$$

After first Iteration:

$$m_1 = 0.075 \quad \checkmark$$

$$c_1 = 0.025 \quad \checkmark$$

Now, y_2 Predicted

$$\hat{y}_2 = m_2 x_1 + c_2$$

$$= 0.075 \times 1 + 0.025$$

$$= \boxed{0.1}$$

$$= m_2 x_2 + c_2$$

$$= 0.075 \times 2 + 0.025$$

$$= \boxed{0.175}$$

$$= m_2 x_3 + c_2$$

$$= 0.075 \times 3 + 0.025$$

$$= \boxed{0.25}$$

$$= m_2 x_4 + c_2$$

$$= 0.075 \times 4 + 0.025$$

$$= \boxed{0.325}$$

Iteration 2

$$m = 0.075 \quad \alpha = 0.01$$

$$c = 0.025$$

Now,

$$m_i = m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$c_i = c - \alpha \frac{\partial}{\partial c} J(m, c)$$

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}) x_i$$

$$= -\frac{1}{4} \left\{ (1 - 0.1) x_1 + (2 - 0.175) x_2 + (3 - 0.25) x_3 + (4 - 0.325) x_4 \right\}$$

$$= -\frac{1}{4} (0.9 + 3.65 + 8.25 + 14.7)$$

$$= -6.875$$

$$\therefore m := 0.075 - (0.01 \times (-6.875))$$

$$= 0.14375 \quad \checkmark$$

again,

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y})$$

$$= -\frac{1}{4} \{ (1 - 0.1) + (2 - 0.175) + (3 - 0.25) + (4 - 0.325) \}$$

$$= -\frac{1}{4} (0.9 + 1.825 + 2.75 + 3.675)$$

$$= -2.2875$$

$$c := 0.025 - (0.01 \times (-2.2875))$$

$$= \boxed{0.047875} \checkmark$$

again practice

$$y = mx + c$$

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

derivation:

$$\text{Square error} = \sum (y_i - \hat{y})^2$$

$$\frac{\partial}{\partial c} J(m, c) = \frac{\partial}{\partial c} \sum (y_i - (mx_i + c))^2$$

$$-2 \sum (y_i - mx_i - c) = 0$$

